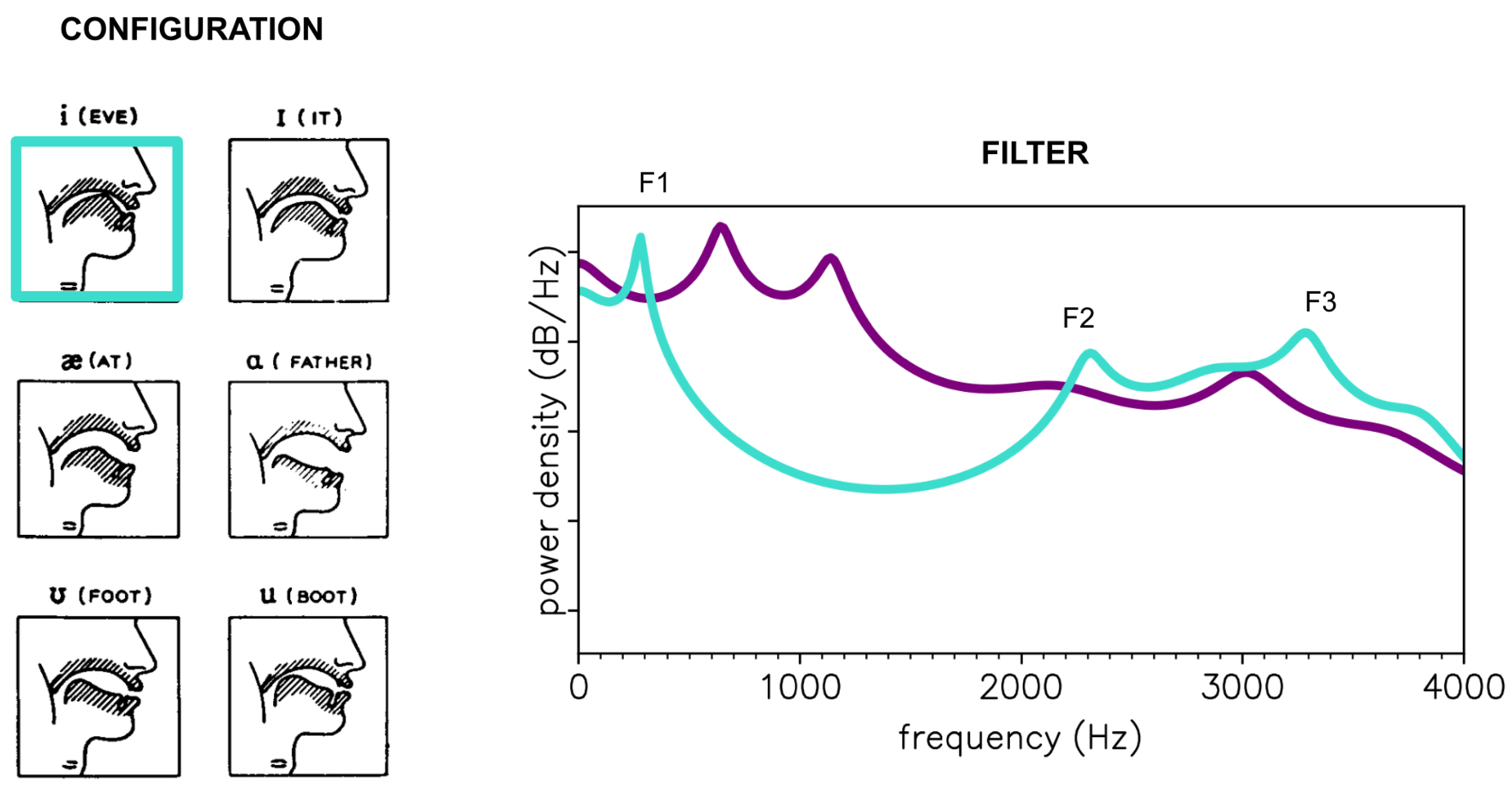
WANTED

A PROBLEM FOR THIS PRIOR

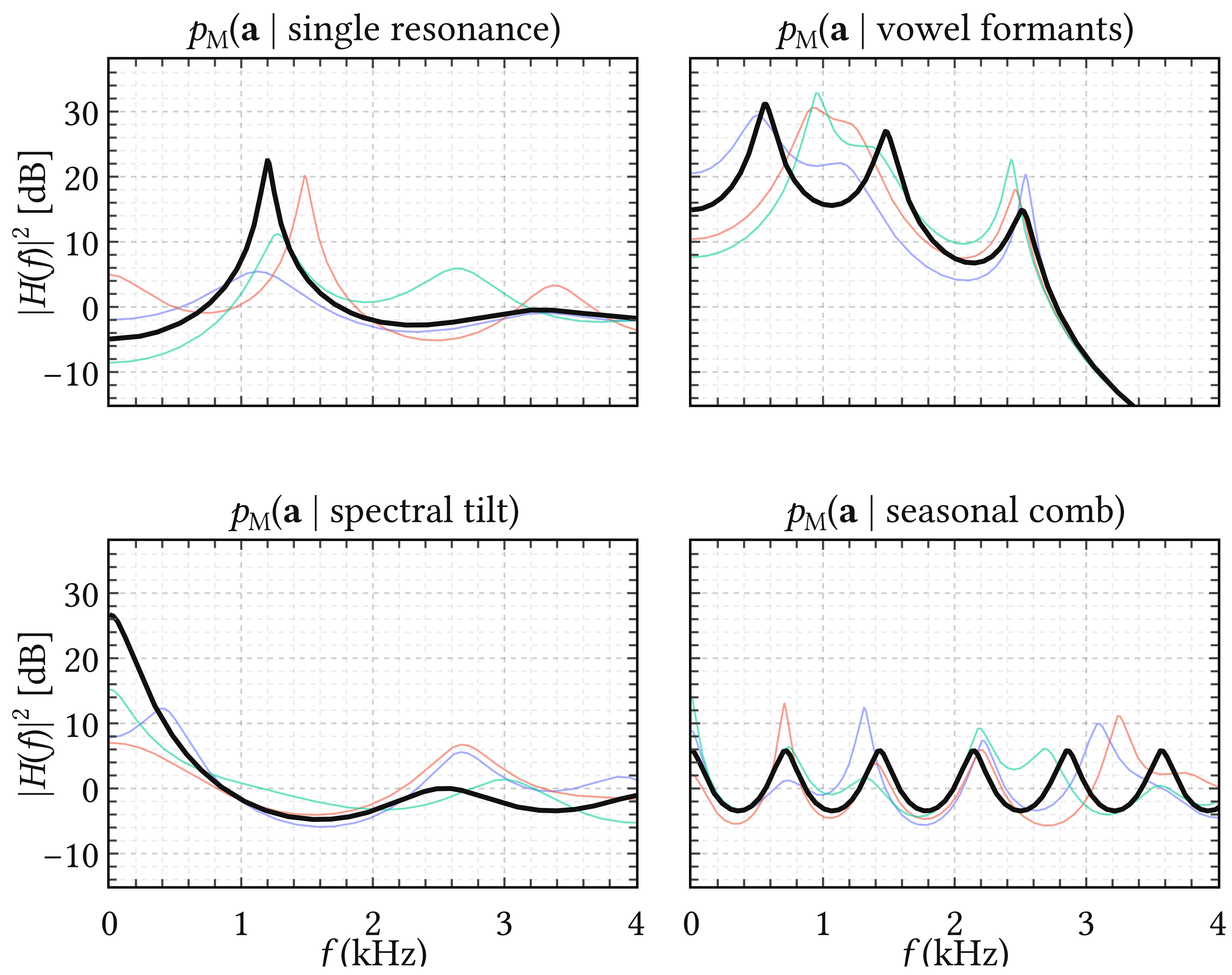
A FILTER IS A POLYNOMIAL

Every resonance you expect is a factor of it.



ONE UPDATE, FOUR BELIEFS

The prior before (grey) and after (black) being told about a resonance, three formants, a tilt, a harmonic comb. One closed-form update, the least change of belief.



OPEN CARDS

Exact, closed form, domain-agnostic. On speech the effect was modest (thesis, ch. 7). Will it matter for *yours*?

THE WHOLE TRICK

AR(4), impose one resonance $Q(z) \mid A(z)$:

$$A(z) = 1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3} - a_4 z^{-4}$$
$$Q(z) = 1 - c z^{-1} + d z^{-2}, \quad c = 2\rho \cos\omega_0, d = \rho^2$$

Divisibility is <u>linear</u> in $\mathbf{a}$:

$$Q \mid A \quad \Longleftrightarrow \quad \mathbf{F}\mathbf{a} + \mathbf{c} = \mathbf{0}, \quad \mathbf{F} = \begin{pmatrix} 1 & 0 & -1/d & -c/d^2 \\ 0 & 1 & c/d & c^2/d^2 - 1/d \end{pmatrix}$$

Impose it *on average*, $\mathbb{E}_{\mathbf{a}}[\mathbf{F}\mathbf{a} + \mathbf{c}] = \mathbf{0}$, by the minimum-$D_{\mathrm{KL}}$ update of $\mathtt{N}(\mathbf{0}, \lambda \mathbf{I})$: a Gaussian prior with the belief in its mean, uncertainty intact.

THE ASK

You fit AR models: vibration, EEG, radar, seismics, econometrics. You know your resonances, tilts and periods. Tell your filter, and see if it pays.

REWARD

a prior with your knowledge in it, and a co-author who checks the math.

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interactive figures